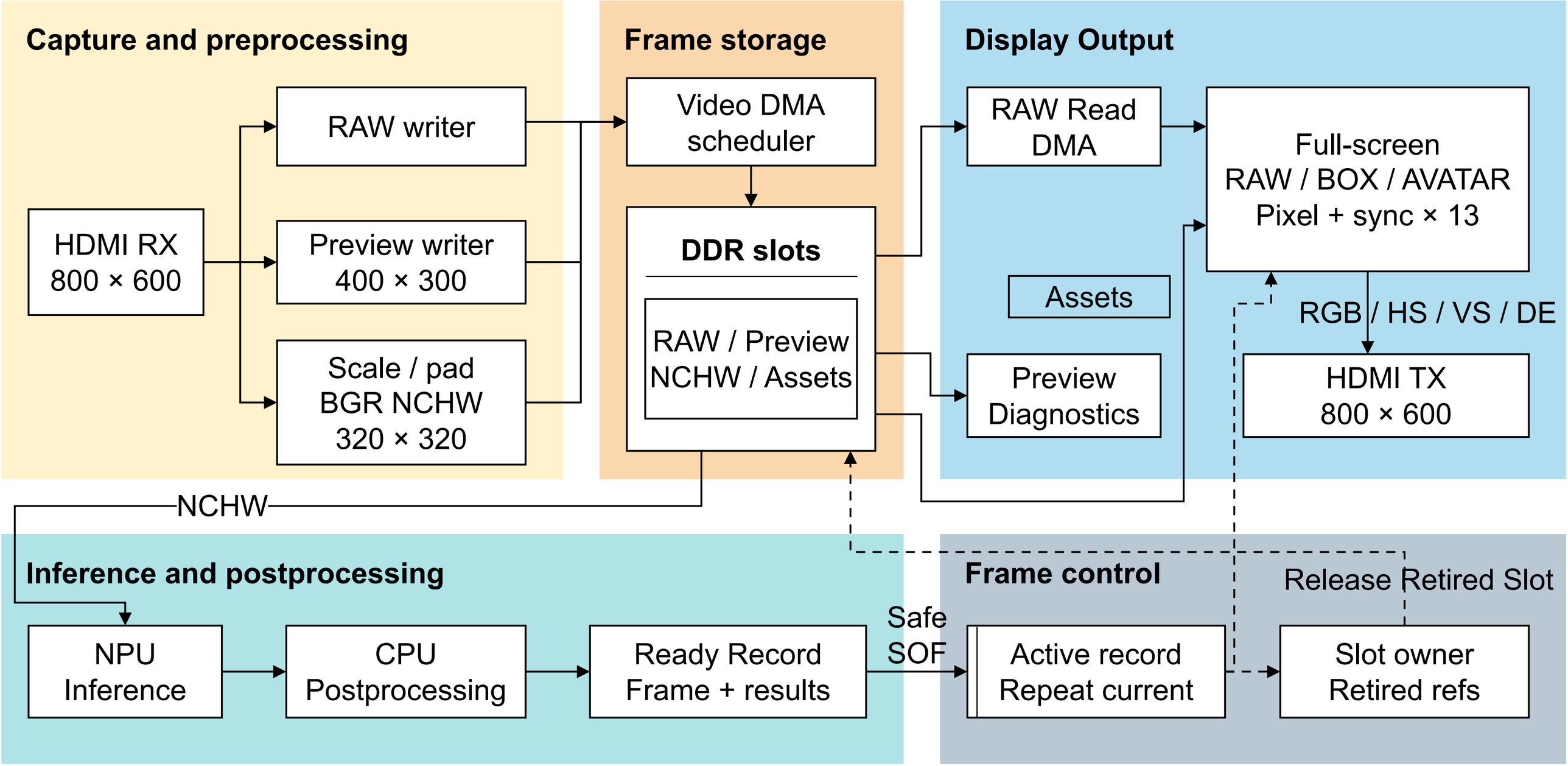
Capture and preprocessing
Frame storage
Display Output
RAW writer
Video DMA scheduler
RAW Read DMA
Full-screen RAW / BOX / AVATAR Pixel + sync × 13
HDMI RX 800 × 600
Preview writer 400 × 300
DDR slots
Assets
RGB / HS / VS / DE
Scale / pad BGR NCHW 320 × 320
RAW / Preview NCHW / Assets
Preview Diagnostics
HDMI TX 800 × 600
NCHW
Inference and postprocessing
Frame control
Release Retired Slot
NPU Inference
CPU Postprocessing
Ready Record Frame + results
Safe SOF
Active record Repeat current
Slot owner Retired refs